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2027 CYCLE

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CUET UG

Mathematics

Numbers, Quantification and Numerical Applications

Modulo, Congruence, Mixtures, Boats, Pipes, Races and Inequalities · Section B2 I · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Modulo arithmetic and congruence	"Modulo Arithmetic: Define modulus of an integer. Apply arithmetic operations using modular arithmetic rules. Congruence Modulo: Define congruence modulo. Apply the definition in various problems." (Section B2 Applied Mathematics Unit I)	Remainder reflexes, reduce-before-multiply discipline, congruence verdicts.
Allegation and mixture	"Allegation and Mixture: Understand the rule of allegation to produce a mixture at a given price. Determine the mean price of a mixture. Apply rule of allegation." (Section B2 Applied Mathematics Unit I)	Criss-cross ratios; verifying a mean price by weighted average.
Applied rate problems	"Boats and Streams: Distinguish between upstream and downstream. Express the problem in the form of an equation. Pipes and Cisterns: Determine the time taken by two or more pipes to fill or empty the tank." (Section B2 Applied Mathematics Unit I)	Building one equation from the story; adding and subtracting rates safely.
Races, games and inequalities	"Races and Games: Compare the performance of two players w.r.t. time, distance. Numerical Inequalities: Describe the basic concepts of numerical inequalities. Understand and write numerical inequalities." (Section B2 Applied Mathematics Unit I, with "Numerical Problems: Solve real life problems mathematically" spanning the unit)	Translating handicaps into speed ratios; flipping inequalities only when required.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The 2026 syllabus opens **Section B2 Applied Mathematics** with this unit — it is the applied-arithmetic engine room where every problem is a story wearing one of a small set of formula coats, so the "which coat?" reflexes below are the whole game.

Modulo Arithmetic — the Mathematics of Remainders

$a \bmod n$ is the remainder when a is divided by n: $17 \bmod 5 = 2$, $24 \bmod 8 = 0$. It is clock arithmetic in the cleanest sense — 3 hours after 10 o'clock reads 1, because $13 \bmod 12 = 1$. The power move for exams is **reduce before you compute**: since $a \bmod n$ only depends on remainders, you may replace each factor by its remainder **before** multiplying, then reduce again. Chasing big powers becomes a small-number game.

THE MODULAR RULES THAT DO THE WORK

$$(a + b) \bmod n = ((a \bmod n) + (b \bmod n)) \bmod n$$

$$(a \cdot b) \bmod n = ((a \bmod n) \cdot (b \bmod n)) \bmod n$$

$a^k \bmod n$: reduce the base, then step the power, reducing each time

WORKED EXAMPLE — BIG PRODUCT, SMALL REMAINDER

Find the remainder when 47×36 is divided by 5, and the remainder when 7^{10} is divided by 5.

Solution. Reduce first: $47 \bmod 5 = 2$ and $36 \bmod 5 = 1$, so the product's remainder is $(2 \cdot 1) \bmod 5 = 2$. For the power: $7 \bmod 5 = 2$, so $7^{10} \bmod 5 = 2^{10} \bmod 5$. Now $2^{10} = 1024$, and $1024 = 204 \cdot 5 + 4$, so the remainder is 4. (Cycle shortcut: $2^n \bmod 5$ cycles $2, 4, 3, 1$; the 10th step lands on 4 — the cycle length 4 divides 8, so position 10 = position 2 in the cycle.) Neither computation ever touched the full product 1692 — the mod engine never needs it.

Congruence Modulo — Same Remainder, Same Class

The notation $a \equiv b \bmod n$ reads "a is congruent to b modulo n" and means exactly one thing: ***n* divides *a* − *b***— the two numbers leave the same remainder. So $29 \equiv 5 \bmod 12$ because $29 - 5 = 24$ is a multiple of 12. Congruence is the syllabus's hidden bridge back to Relations: it is **reflexive** ($a - a = 0$), **symmetric** (reverse the subtraction), and **transitive** (chain the divisibility) — a full equivalence relation, sorting the integers into n remainder classes.

- **Rewrite before computing:** $29 \equiv 5 \bmod 12$ lets you replace 29 by 5 inside any sum or product mod 12 — the whole toolkit of the previous section in one notation.
- **Calendar questions are congruence questions:** "what day is it 100 days after a Tuesday?" is $100 \bmod 7 = 2$ days after — Thursday.
- **Verdict questions:** "$35 \equiv 11 \bmod 8$?" — check $35 - 11 = 24$, and 24 is a multiple of 8 ✓. One subtraction, one division, done.

Allegation and Mixture — the Criss-Cross Rule

Mix two ingredients priced differently; what should the blend cost, or what ratio gives a target price? The **rule of allegation** answers with a criss-cross subtraction: if the cheap item costs c, the dear item costs d, and the mixture's mean price is m (which must sit between them), then

$$\text{ratio of cheap} : \text{ratio of dear} = (d - m) : (m - c)$$

— each side gets the **opposite** gap. The rule is just a weighted average written diagonally, and the verification habit (recompute the weighted mean) takes one line and catches every reversed-ratio slip.

WORKED EXAMPLE — BLENDING RICE TO A TARGET PRICE

In what ratio must rice at ₹30 per kg be mixed with rice at ₹45 per kg so that the mixture costs ₹36 per kg?

Solution. Cheap $c = 30$, dear $d = 45$, mean $m = 36$. Criss-cross:

$$\text{cheap} : \text{dear} = (45 - 36) : (36 - 30) = 9 : 6 = 3 : 2.$$

Verify: $\frac{3\cdot30+2\cdot45}{3+2} = \frac{90+90}{5} = \frac{180}{5} = 36$ ✓. Three parts cheap to two parts dear. The most common error — writing $(36 - 30) : (45 - 36)$ — inverts the ratio; the one-line verification catches it before the option list does.

Boats and Streams — Two Speeds, One Boat

A boat in moving water travels at its **still-water speed** b boosted or dragged by the **stream speed** s: downstream $d = b + s$, upstream $u = b - s$. Half the questions hand you u and d and ask for the pair — the rescue formulas swap the algebra for arithmetic:

$$b = \frac{u + d}{2}, \qquad s = \frac{d - u}{2}$$

("add for the boat, subtract for the stream"). Time questions then run on $\text{time} = \frac{\text{distance}}{\text{speed}}$ with the correct speed chosen by direction.

WORKED EXAMPLE — RECOVER THE BOAT AND THE STREAM

A boat covers 24 km downstream in 2 hours and the same distance upstream in 3 hours. Find the boat's still-water speed and the stream's speed.

Solution. Downstream speed: $\frac{24}{2} = 12$ km/h. Upstream: $\frac{24}{3} = 8$ km/h.

$$b = \frac{12 + 8}{2} = 10 \text{ km/h}, \quad s = \frac{12 - 8}{2} = 2 \text{ km/h}.$$

Sanity check: downstream should be faster — $12 > 8$ ✓ — and the stream pushing at 2 moves 10 to 12 and drags it to 8, exactly as booked.

Pipes and Cisterns — Rates Add, Empties Subtract

Every pipe contributes a **rate** measured in tank per hour: a pipe that fills in 6 hours fills $\frac{1}{6}$ of the tank per hour. Fillers add their rates; emptying pipes **subtract** theirs; the net rate's reciprocal is the total time. Part-filled checkpoints ("how much after 1 hour?") are just rate × time.

WORKED EXAMPLE — ONE EMPTIER AMONG TWO FILLERS

Pipe A fills a tank in 6 hours, pipe B in 4 hours, and pipe C empties it in 12 hours. All three open together — how long to fill the tank?

Solution. Rates per hour: $A = \frac{1}{6}$, $B = \frac{1}{4}$, $C = -\frac{1}{12}$. Net:

$$\frac{1}{6} + \frac{1}{4} - \frac{1}{12} = \frac{2}{12} + \frac{3}{12} - \frac{1}{12} = \frac{4}{12} = \frac{1}{3} \text{ tank per hour.}$$

Time $= \frac{1}{\frac{1}{3}} = 3$ hours. The emptier genuinely slowed things — without C the pair would need $\frac{1}{\frac{2}{12}+\frac{3}{12}} = \frac{12}{5} = 2.4$ hours; C added 0.6 hours of work. Every pipes question is this one arithmetic in different clothes.

Races and Games — Reading the Handicap

"A gives B 10 metres in a 100 metre race" means: **when A completes 100 m, B has covered 90 m**. The handicap converts directly into a speed ratio, because both ran for the same time — distances in equal times carry the speed ratio. "A gives B 10 seconds" is the time-handicap twin: B may start 10 s late (or A waits), and the shared clock does the comparing. Games-of-score questions reduce to the same idea with points instead of metres: convert the statement into equal-time (or equal-round) totals, then form the ratio.

WORKED EXAMPLE — FROM HANDICAP TO SPEED RATIO

In a 100 m race, A beats B by 10 metres; in a 160 m race, B beats C by 4 metres. By how many metres can A beat C in a 100 m race?

Solution. Chain the ratios. A : B = 100 : 90 = 10 : 9. B : C = 160 : 156 = 40 : 39. Multiply: A : C = (10 · 40) : (9 · 39) = 400 : 351. So when A runs 100 m, C runs $\frac{351}{400} \cdot 100 = 87.75$ m — A beats C by $100 - 87.75 = 12.25$ metres. Two handicaps, one chained ratio, no hero algebra.

Numerical Inequalities — the Flip Rule and Its Limits

Numerical inequalities compare sizes, and one property does nearly all the work: **multiplying or dividing both sides by a negative number flips the direction**. Adding or subtracting anything flips nothing; multiplying by a positive flips nothing; multiplying by a negative flips everything. For positive numbers, taking reciprocals also flips: $a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$ — which is why the fastest runner has the smallest time.

WORKED EXAMPLE — TWO VERDICTS, TWO RULES

If $3x > 12$ and $-2y < -10$, what can be said about x and y?

Solution. Divide the first by the positive 3: $x > 4$ — no flip. Divide the second by the negative -2: the inequality **flips**, giving $y > 5$. So x exceeds 4 and y exceeds 5 — and any option claiming $y < 5$ fell for the un-flipped division. Say the sign of the divisor out loud every time; it is the entire topic.

EVERYDAY EXAMPLE — THE BUILDING WITH REPEAT FLOORS

Modulo arithmetic is a building whose floor numbers repeat every n levels: climb 13 floors in a 5-floor cycle and you arrive at floor 3 — $13 \bmod 5$. The congruence sign is the lift-attendant's statement "floor 29 and floor 5 are the same place in this building" — true precisely because the difference, 24, is whole cycles. Allegation is the **grocery blend**: to hit a shelf price between basmati and generic rice, lean the scoop toward whichever grain's price gap is smaller on the opposite side. Pipes are the **bathtub with the drain open**: what matters is the net inflow per minute, fillers minus drain. And a race handicap is the **head start on the track** — the clock never stops, so the distance gap at the tape **is** the speed ratio.

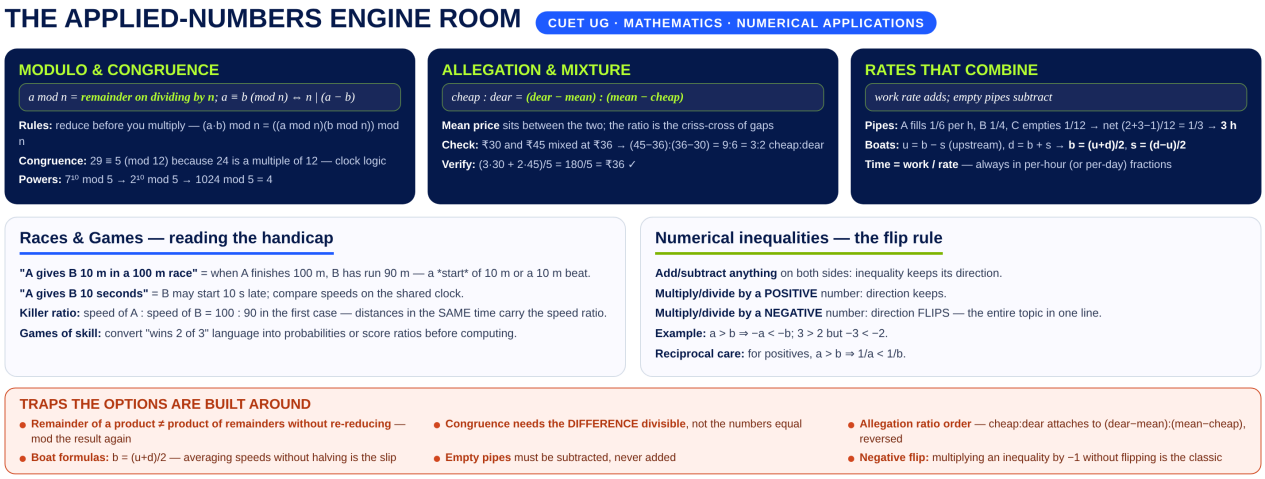


Figure 1: The engine room on one board: the mod engine, the allegation criss-cross, the rate-combining cards — and the traps under each.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — NUMBERS AND NUMERICAL APPLICATIONS

Formula / rule	Letters mean	Fire it when
$a \bmod n$ = remainder on division by n	clock arithmetic	all remainder questions

Formula / rule	Letters mean	Fire it when
$(a \cdot b) \bmod n = ((a \bmod n)(b \bmod n)) \bmod n$	reduce, multiply, reduce	big products
$a \equiv b \bmod n \Leftrightarrow n \mid (a-b)$	same remainder class	congruence verdicts
Congruence is reflexive, symmetric, transitive	equivalence relation	definition MCQs
Allegation: cheap : dear $= (d-m) : (m-c)$	opposite gaps	mixture ratios
Mean price $= \frac{\text{cheap} \cdot w_1 + \text{dear} \cdot w_2}{w_1 + w_2}$	weighted average	verification
Downstream $d = b + s$; upstream $u = b - s$	boost and drag	boat setups
Boat $b = \frac{u+d}{2}$; stream $s = \frac{d-u}{2}$	add for boat, subtract for stream	rescue pair
Pipe rate $= \frac{1}{\text{hours taken}}$ per hour	fillers add, empties subtract	tanks
Time to fill $= \frac{1}{\text{net rate}}$	reciprocal of the sum	combined pipes
"A gives B x m in an N m race" $\rightarrow$ speeds $N : (N-x)$	equal time, ratio of distances	race handicaps
Multiply by a negative $\rightarrow$ inequality flips	the flip rule	inequality algebra
Positive numbers: $a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$	reciprocal flip	rate-vs-time comparisons

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — NUMERICAL APPLICATIONS

The mistake	What is right	The trap it sets
Multiplying full numbers then reducing	Reduce factors first, product again	Unwieldy arithmetic and time loss
Congruence checked by "same last digit"	Check $n \mid (a-b)$ properly	Coincidental digit matches
Allegation ratio attached to the wrong side	Cheap gets $(d-m)$, dear gets $(m-c)$	Inverted-ratio options

The mistake	What is right	The trap it sets
Boat formulas averaged without halving	$b = \frac{u+d}{2}$, $s = \frac{d-u}{2}$	Doubled speed answers
Empty pipes added as fillers	Their rates subtract	Tanks that never fill in the options
Speeds compared across different times	Equal-time distances carry the ratio	Race answers off by a margin
Division by a negative without flipping	Flip the inequality, always	$y < 5$-style wrong verdicts
Cycle miscount in powers	Position = power mod cycle length	Off-by-one remainders

Sixty-Second Self-Test

Statement	Verdict
$23 \bmod 6 = 5$.	True — 18 fits, remainder 5
$41 \equiv 9 \bmod 8$.	True — difference 32 is a multiple of 8
Mixing ₹20 and ₹50 items to average ₹40 needs more of the dear item.	True — mean closer to 50
Boat 5 km/h, stream 1 km/h: upstream 6 km/h.	False — upstream drags to 4
An emptying pipe's rate is added like a filler.	False — it subtracts
If $a > b > 0$ then $\frac{1}{a} > \frac{1}{b}$.	False — reciprocals flip

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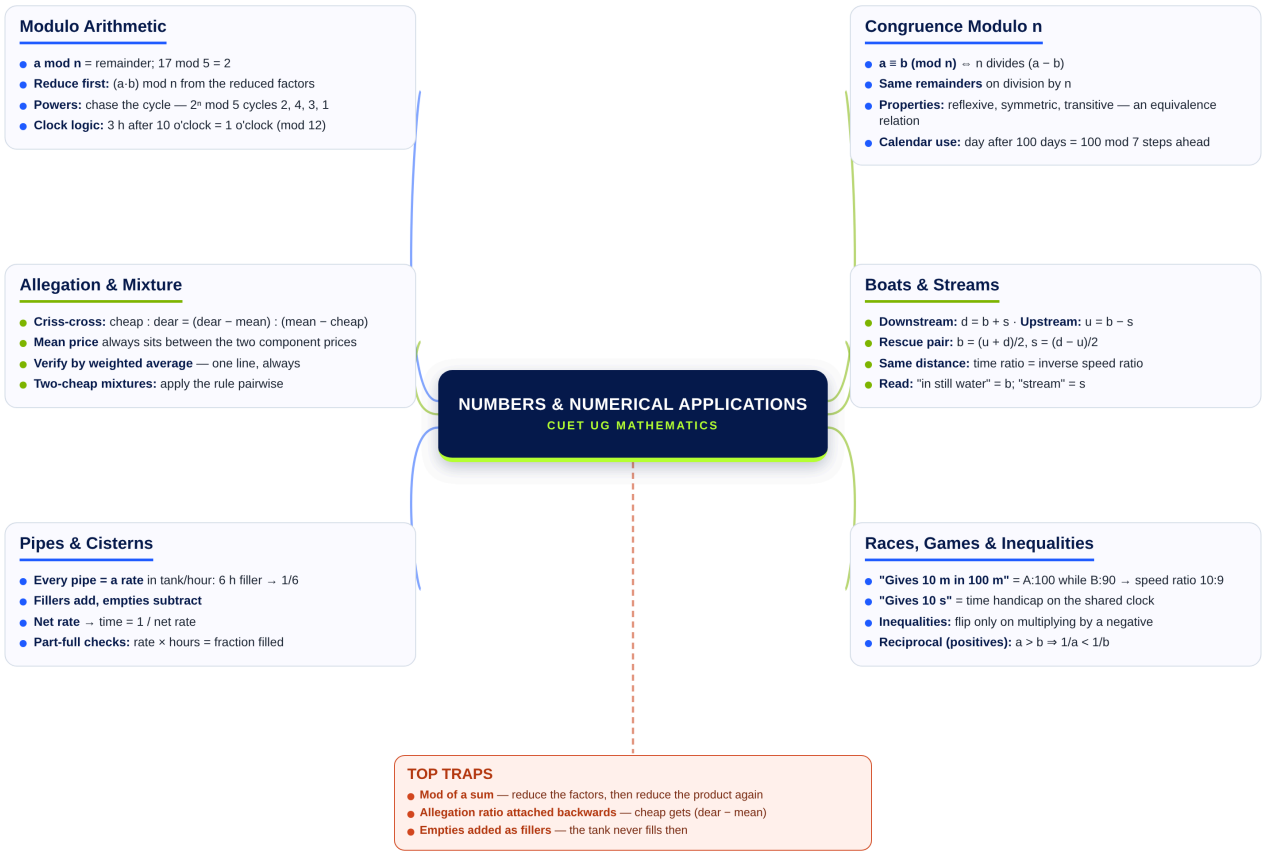


Figure 2: The unit in one look: modulo, congruence, mixtures, boats, pipes, races and inequalities — with the traps between them.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Name the coat — mod, mixture, boat, pipe, race or inequality — before computing.

Q1. The remainder when 43×67 is divided by 5 is:

(A) 0 (B) 1
(C) 2 (D) 3

Q2. If $7 \equiv 2 \bmod 5$, then the remainder when 7^4 is divided by 5 is:

(A) 2 (B) 3
(C) 4 (D) 1

Q3. Which statement is correct?

(A) $17 \equiv 3 \bmod 5$ (B) $17 \equiv 4 \bmod 5$
(C) $17 \equiv 1 \bmod 5$ (D) $17 \equiv 2 \bmod 5$

Q4. In what ratio must tea at ₹24 per kg be mixed with tea at ₹42 per kg to produce a mixture worth ₹30 per kg?

(A) 2 : 1 **(B)** 1 : 2
(C) 3 : 2 **(D)** 2 : 3

Q5. A boat's downstream speed is 14 km/h and its upstream speed is 6 km/h. The speed of the stream is:

(A) 10 km/h **(B)** 8 km/h
(C) 4 km/h **(D)** 2 km/h

Q6. Pipe A fills a tank in 3 hours and pipe B fills it in 6 hours. Together they fill it in:

(A) 4 hours **(B)** 4.5 hours
(C) 2 hours **(D)** 9 hours

Q7. In a 100 m race, A beats B by 25 metres. The ratio of their speeds is:

(A) 3 : 1 **(B)** 4 : 3
(C) 5 : 4 **(D)** 25 : 24

Q8. If $-3x < -12$, then:

(A) $x > 4$ **(B)** $x < 4$
(C) $x < -4$ **(D)** $x > -4$

Q9. A tank is filled by one pipe in 2 hours and emptied by another in 4 hours. With both open, the tank fills in:

(A) 3 hours **(B)** 2 hours
(C) 1 hour **(D)** 4 hours

Q10. 100 days after a Tuesday, the day of the week is:

(A) Friday **(B)** Saturday
(C) Thursday **(D)** Sunday

Detailed Solutions

Sol. Q1. Reduce first: $43 \bmod 5 = 3$ and $67 \bmod 5 = 2$. Product remainder: $(3 \cdot 2) \bmod 5 = 6 \bmod 5 = 1$. The full product 2881 confirms it: $2881 = 576 \cdot 5 + 1$.

Answer: (B)

Sol. Q2. Replace the base: $7^4 \equiv 2^4 = 16 \bmod 5 = 1$. Cycle check: $2^n \bmod 5$ cycles 2, 4, 3, 1 — position 4 lands on 1 ✓.

Answer: (D)

Sol. Q3. $17 = 3 \cdot 5 + 2$, so $17 \bmod 5 = 2$ and $17 \equiv 2 \bmod 5$. Equivalently $17 - 2 = 15$ is a multiple of 5; none of the other differences are.

Answer: (D)

Sol. Q4. Criss-cross with $c = 24$, $d = 42$, $m = 30$: cheap : dear $= (42 - 30) : (30 - 24) = 12 : 6 = 2 : 1$. Verify: $\frac{2 \cdot 24 + 1 \cdot 42}{3} = \frac{90}{3} = 30$ ✓.

Answer: (A)

Sol. Q5. Rescue formula: $s = \frac{d-u}{2} = \frac{14-6}{2} = 4$ "km/h" (and the boat itself: $b = 10$). The average-without-halving slip gives the 10 distractor.

Answer: (C)

Sol. Q6. Rates: $\frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$ tank per hour, so the pair fills in 2 hours. Option (D) added hours instead of rates.

Answer: (C)

Sol. Q7. When A finishes 100 m, B has run 75 m. Equal time, so speeds: A : B $= 100 : 75 = 4 : 3$. The 25-metre handicap is the ratio's whole content.

Answer: (B)

Sol. Q8. Divide by -3 — negative divisor, so the inequality flips: $x > 4$. The un-flipped $x < 4$ is the trap this exact question exists to catch.

Answer: (A)

Sol. Q9. Rates: $+\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ tank per hour net, so the tank fills in 4 hours. The emptier halves the effective speed.

Answer: (D)

Sol. Q10. $100 \bmod 7 = 2$ (14 weeks = 98 days), so the day advances by 2: Tuesday → Wednesday → Thursday. Congruence dressed as a calendar.

Answer: (C)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(D) in Q1	Added remainders but forgot the final reduction	$(3 + 2) \bmod 5$ needs one more mod pass — always reduce again

If you picked this	The thinking slip	The lasting fix
(C) in Q5	Subtracted without halving	The stream is half the gap: $\frac{14-6}{2}$
(D) in Q6	Added the hours $3 + 6 = 9$	Add RATES $(1/3 + 1/6)$, then invert — never the hours
(A) in Q8	Divided by a negative without flipping	Say the divisor's sign aloud; negative = flip
(C) in Q9	Treated 2 h filler as overpowering 4 h emptier	Net rate arithmetic: $1/2 - 1/4 = 1/4$, then invert
(B) in Q10	Counted 100 mod 7 as 3	$7 \cdot 14 = 98$; remainder is 2 — verify the multiple first

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